

DECODE Scoring and Strategy Background

What Do Teams Need to Know Before Choosing a Strategy?

In DECODE, two alliances compete at the same time. Each alliance has two robots. Robots try to score game pieces, build patterns, avoid penalties, and return to their base before time runs out.

This course uses a simplified version of the full FTC challenge. Your team will still use the real ideas of engineering strategy: choosing what to attempt, designing under constraints, testing with evidence, and explaining why your choices make sense.

The Big Idea

A strong robot is not always the robot that tries to do the most things. A strong robot is one that can do useful things reliably.

For this course, teams should think about four strategy questions:

- How will we score balls or artifacts?
- Where will we launch from?
- What can we do in autonomous?
- How will we help both alliance robots park or return to base?

Course Constraints

These constraints shape your strategy.

Constraint	What It Means
No intake	Your robot will not collect balls from the field. A human operator will load balls into the robot.
Hopper limit goal	Your hopper should be designed to carry up to three balls.
No AprilTag reading	Your robot will not use camera vision to identify field targets.
No distance sensors	Your robot will not automatically measure exact distance from the goal.
Limited build time	A simple reliable robot may outperform an unfinished complex robot.
Shared field space	Your robot should leave room for an alliance partner to move, score, and park.

These are not failures. They are engineering constraints. Good teams make smart choices inside real limits.

Match Periods

FTC matches have two main periods.

Period	What Happens
Autonomous	Robots run pre-programmed instructions without driver control.
TeleOp	Human drivers control the robots.

In the official DECODE game, autonomous can include moving off the launch line, scoring artifacts, and building a pattern. In this course, every team should try for a simple autonomous movement. Some teams may challenge themselves to attempt autonomous scoring if their robot is ready.

Scoring Ideas That Matter for This Course

The official DECODE game includes several ways to earn points. Your team does not need to master every official rule to make a strong strategy, but you do need to understand the parts that affect your design.

Scoring Idea	Why It Matters
Scoring artifacts or balls	Your launcher, hopper, driver control, and launch site all affect scoring.
Leaving the launch line in autonomous	A simple autonomous movement can create reliable early points.
Returning to base	Parking or returning to base can be a valuable endgame goal.
Two robots fully returned to base	Alliances can earn an additional bonus when both robots fully fit in base.
Ranking points	Some achievements can matter beyond match points because they help teams rank higher.
Penalties	A risky strategy can lose value if it creates penalties.

Why Parking and Robot Size Matter

In DECODE, both robots may need to return to the alliance base at the end of the match. If both robots can fully fit, the alliance can earn an additional bonus.

This makes robot size a strategy decision. A team might choose:

- a narrow robot that leaves room for an alliance partner,
- a compact robot that can park quickly,
- a taller or unusual robot shape that tries to share space differently,
- or a wider robot that is easier to build but may make parking harder.

The unusual choice is not automatically better. A narrow or tall robot may create problems with balance, hopper space, launcher placement, battery access, and driver control. The team should only choose that path if the parking benefit is worth the trade-off.

Human Operator Loading

Because this course does not include an intake, your human operator matters.

Your team should decide:

- where the human operator loads balls,
- how the robot lines up for loading,
- how many balls the hopper can safely hold,
- how the driver knows the robot is ready to leave,

- and how the team avoids unsafe or illegal loading.

A fast launcher does not help if the robot takes too long to load or if the hopper jams.

Launch Site Choice

Because your robot will not use AprilTag reading or distance sensors, your team needs a repeatable launch site.

A good launch site should be:

- legal,
- reachable,
- easy for the driver to recognize,
- far enough from traffic or defense,
- compatible with the launcher angle and power,
- and respectful of the alliance partner's path.

Your team should choose one first launch site and one backup launch site.

Autonomous Goals

Every team should try to create at least one autonomous program that moves the robot predictably.

Possible autonomous goals:

Goal	Difficulty	Why a Team Might Choose It
Drive off the launch line or starting area	Easier	Reliable first autonomous success.
Drive to a useful position	Medium	Sets up TeleOp or avoids traffic.
Park or move toward base	Medium	Supports a simple endgame or field-position plan.
Score autonomously	Hard	Strong challenge if the launcher and starting setup are consistent.

Autonomous scoring is exciting, but it is risky without sensors. Small changes in starting position, battery level, wheel traction, or ball placement can change the result.

Penalty Avoidance

Avoiding penalties is part of strategy. A team can lose the value of a good scoring plan by breaking rules, blocking illegally, loading unsafely, contacting where contact is not allowed, or continuing an action after time expires.

Teams should ask:

- What are the easiest penalties for us to accidentally commit?
- What should the driver avoid?
- What should the human operator avoid?
- What should the coach remind the team about?
- What part of our strategy needs a rules check before we rely on it?

Strategy Trade-Offs

Good strategy means choosing what not to do.

If We Prioritize...	It Helps Us...	But It May Make It Harder To...
Reliable three-ball scoring	Build and practice one clear plan	Score more than one cycle
Multiple scoring cycles	Increase scoring potential	Stay accurate, avoid traffic, and avoid penalties
A fixed launch site	Aim more consistently	Recover if the site is blocked
Narrow parking-friendly robot	Fit with an alliance partner	Fit the hopper, launcher, battery, and electronics
Simple autonomous movement	Earn reliable early value	Attempt autonomous scoring
Autonomous scoring	Create strong challenge evidence	Finish other systems on time
Penalty avoidance	Protect the score	Drive aggressively or attempt risky plays

Evidence Teams Should Collect

A good strategy should produce evidence. During testing, collect information such as:

- number of balls scored out of three,
- time needed to load, drive, aim, and shoot,
- number of complete scoring cycles in a match-length test,
- accuracy from the first launch site and backup launch site,
- whether two robots can fully fit in base,
- autonomous success rate across several trials,
- driver feedback about control and aiming,
- and rule or penalty concerns noticed during practice.

Main Takeaway

Your team does not need the most complicated robot. Your team needs a strategy that fits your materials, time, drivers, human operator, alliance partner, and evidence from testing.

The best engineering explanation sounds like this:

We chose this strategy because it gave us the best chance to score reliably, avoid penalties, and support our alliance. We accepted this trade-off, and our testing evidence showed this was the right next step.

Game Strategy and Goals Menu

What Should Our Robot Try to Do in a Match?

Graduate Expectation Tags

- **[Red: Critical Thinking & Problem Solving]** We compare scoring, parking, reliability, and penalties before choosing a strategy.
- **[Blue: Effective Communication]** We explain our match plan clearly so drivers, human operators, builders, programmers, and presenters understand it.
- **[Purple: Curiosity & Creativity]** We imagine different ways to score, park, and work with an alliance partner.
- **[Green: Personal Development]** We choose a realistic goal based on time, materials, skill, and evidence.
- **[Orange: Cultural Understanding & Civic Engagement]** We think about alliance teamwork, field space, and responsible competition.

Decision Being Made

Your team is choosing its first match strategy. A strategy is not just what the robot can do. It is what the team will practice, test, explain, and improve.

In this course, your robot will not use an intake. A human operator will load balls into the robot. Your team will design a hopper that can carry up to three balls and a launching system that can score those balls.

Your team also has limited time and materials. You do not have sensors for reading AprilTags or measuring exact distance, so aiming and driving will depend on driver practice, field marks, repeatable starting positions, and simple autonomous movements.

Background Sources

Use ideas from:

- **What Is FIRST Tech Challenge?**
- **Robotics Safety Norms**
- **Borrowed Equipment and Community Trust**
- **Chassis Shape Menu**
- **Wheel Selection Menu**
- **Motor Placement Menu**
- **Electronics and Battery Placement Menu**
- **DECODE Scoring and Strategy Background**
- drive practice, field observations, and early chassis tests
- the current game rules or course scoring summary

Why This Decision Matters

Teams often want to build every possible scoring feature. In a short course, that can make the robot late, fragile, or hard to drive. A strong strategy helps the team decide what matters most.

A good strategy answers questions like:

- Where should the robot launch from?
- How many scoring trips should the team attempt?
- How will the human operator load balls quickly and safely?
- How will the robot avoid penalties?
- How will the robot leave room for an alliance partner?
- Should the team prioritize reliable parking, scoring, or an ambitious challenge?
- What evidence will show whether the strategy is working?

There is no perfect strategy. Your job is to choose a strategy that your team can build, test, drive, and explain with evidence.

Important Course Constraints

These are not menu options. They are constraints your team must design around.

Constraint	What It Means for Strategy
No intake	The robot will not collect balls from the field. A human operator will load balls.
Three-ball maximum hopper goal	The hopper should hold up to three balls so the robot can attempt a scoring run after loading.
No AprilTag or distance sensors	Aiming and autonomous movement must rely on driver judgment, field marks, timing, encoder estimates, or repeatable starting positions.
Limited build time	A reliable simple strategy may score more than a complex unfinished strategy.
Shared field space	The robot should avoid blocking an alliance partner unless that is part of an agreed strategy.
Penalties matter	Avoiding penalties can be as important as scoring more points.

Menu Options

Option A: Reliable Three-Ball Scoring Run

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The robot is designed to load up to three balls, drive to a chosen launch site, aim by driver judgment or field marks, and score the three balls as reliably as possible.

What it helps: This keeps the strategy focused. The team can practice one repeatable scoring path and collect evidence about accuracy, cycle time, and driver control.

Best for: Teams that want a clear first scoring goal and a robot that is easier to build, test, and present.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Possible trade-offs: The team may score fewer total balls than a faster cycle strategy. If the launch site is blocked or defended, the robot needs a backup plan.

Evidence we might watch: How many of three balls score from the chosen site? How long does one loading-driving-shooting cycle take? Can different drivers repeat the path?

Option B: Two or Three Short Scoring Cycles

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The team tries to make multiple trips: load, drive, score, return, reload, and score again. The goal is to score more than the first three balls.

What it helps: This can increase scoring if the robot drives reliably, the human operator loads quickly, and the launch site is easy to reach.

Best for: Teams with a reliable chassis, quick loading plan, and drivers who can repeat a route without losing control.

Ease of use: Medium to hard

Likelihood of success: Medium

Time to implement: Moderate to slow

Possible trade-offs: More cycles create more chances for missed shots, traffic, defense, loading delays, or penalties. The team may spend too much time chasing speed before the first scoring cycle is reliable.

Evidence we might watch: Can the robot complete two full cycles before time runs out? Does the second cycle score as reliably as the first? Where does the team lose the most time?

Option C: Specific Launch Site Strategy

Graduate Expectation Connection: [Blue: Effective Communication] and [Red: Critical Thinking & Problem Solving]

What it means: The team chooses one legal launch site and designs driving practice, aiming, hopper position, and shooter angle around that location.

What it helps: A known launch site can make aiming more repeatable. It also helps the team communicate with an alliance partner about field space.

Best for: Teams that need consistency because they are aiming without distance sensors or AprilTag reading.

Ease of use: Medium

Likelihood of success: High if the site is legal, reachable, and repeatable

Time to implement: Fast to moderate

Possible trade-offs: The site may be blocked by another robot, a field object, or defense. A strategy built around one spot needs a backup site or recovery plan.

Evidence we might watch: Can the robot reach the site without hitting field elements? Does the robot score more accurately from that site than from other sites? Can the alliance partner still move and park?

Option D: Alliance-Friendly Parking Strategy

Graduate Expectation Connection: [Orange: Cultural Understanding & Civic Engagement]

What it means: The robot is designed and driven so both alliance robots can fully park or fit in the required end area when the match ends.

What it helps: This may earn extra endgame, ranking, or advancement value if the current rules reward both alliance robots parking. It also shows strong alliance awareness.

Best for: Teams that want a reliable endgame contribution and are willing to design around shared space.

Ease of use: Medium

Likelihood of success: Medium to high if practiced

Time to implement: Moderate

Possible trade-offs: A narrow robot or unusual shape may leave less room for hopper, launcher, battery, electronics, or stability. A tall robot that allows another robot underneath may be risky, harder to build, and harder to keep stable.

Evidence we might watch: Can two robots fully fit in the parking area? Can the robot park quickly after scoring? Does the robot remain stable if it is tall or narrow?

Option E: Simple Autonomous Mobility

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: The robot has at least one autonomous program that drives away from the starting area or off the back in a predictable way.

What it helps: This gives the team a realistic autonomous goal. It can be tested early and improved without requiring sensors or complex aiming.

Best for: Teams that need a dependable autonomous routine before attempting autonomous scoring.

Ease of use: Easy to medium

Likelihood of success: High with testing

Time to implement: Fast to moderate

Possible trade-offs: This may score fewer autonomous points than a scoring routine, but it is more likely to work consistently.

Evidence we might watch: Does the robot end in the correct area three times in a row? Does battery level or starting position change the result? Can the team explain the program?

Option F: Autonomous Scoring Challenge

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The team attempts to score during autonomous using a repeatable starting position, timed movement, motor power, encoder estimates, or a fixed launch setup.

What it helps: This is an ambitious challenge that can create strong portfolio evidence if the team tests carefully and explains what worked or failed.

Best for: Teams that already have reliable basic autonomous movement and a launcher that scores consistently from a known setup.

Ease of use: Hard

Likelihood of success: Risky

Time to implement: Slow

Possible trade-offs: Autonomous scoring may consume time needed for driver practice, hopper reliability, or basic scoring. Without sensors, small changes in starting position may cause misses.

Evidence we might watch: How often does autonomous scoring work from the same starting setup? What variable causes misses: starting angle, launch speed, battery level, ball position, or robot drift?

Option G: Penalty-Avoidance Strategy

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: The team makes avoiding penalties a major part of the match plan. Drivers, human operators, and coaches practice legal loading, legal launching, safe contact, and field awareness.

What it helps: Avoiding penalties protects the team's score and shows discipline. A clean, reliable match may beat a risky match with more attempted scoring.

Best for: All teams, especially teams with new drivers or a robot that will operate near other robots.

Ease of use: Medium

Likelihood of success: High with practice

Time to implement: Fast to moderate

Possible trade-offs: The team may drive more cautiously and attempt fewer shots while learning the rules.

Evidence we might watch: How many practice runs are completed without rule warnings? Can the driver and human operator explain the riskiest penalty situations?

Strategy Comparison Table

Strategy	Best Feature	Main Risk	Good For
Reliable three-ball scoring run	Clear and testable	May limit total scoring	First scoring plan
Two or three short scoring cycles	Higher scoring potential	More chances for delays and mistakes	Fast, reliable drive teams
Specific launch site	Repeatable aiming	Site may be blocked	Sensor-free scoring
Alliance-friendly parking	Helps alliance endgame	May constrain robot shape	Teams designing for shared space
Simple autonomous mobility	Reliable first autonomous	Lower ambition	Early autonomous success
Autonomous scoring challenge	Strong stretch evidence	Risky without sensors	Advanced testing teams
Penalty avoidance	Protects score	May feel cautious	All teams

Strategy Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Main scoring strategy				
Launch site				
Number of scoring cycles				
Parking plan				
Autonomous goal				
Penalty-avoidance focus				

Launch Site Planning

Choose one first launch site and one backup launch site.

Launch Site	Why It Might Work	What Could Block It	Evidence to Test
First choice			
Backup choice			

Before finalizing a launch site, check:

- Is the site legal under the current rules?
- Can the robot reach the site safely?
- Can the human operator load the robot before it drives there?
- Can the driver aim from that site without sensors?
- Can an alliance partner still move, score, or park?
- What should the robot do if the site is blocked?

Must / Should / Could

Level	Expectation
Must Do	Choose a main scoring strategy, one legal launch site, and one autonomous mobility goal.
Should Do	Explain the main trade-off in the strategy and identify evidence the team will collect during testing.
Could Do	Compare two strategies using cycle time, scoring accuracy, parking space, penalty risk, or alliance impact.

SLD Connection

Main SLD Prompt

Should our team prioritize reliable three-ball scoring, multiple scoring cycles, alliance-friendly parking, or autonomous challenge points?

Supporting Questions

- What can our robot realistically do with no intake?
- How many balls can we score reliably from one loading cycle?
- Where is our best legal launch site?
- What launch site still leaves room for an alliance partner?
- Should our robot shape help two alliance robots park?
- What is the safest first autonomous goal?
- When would autonomous scoring be worth the extra risk?
- What penalties are most important for us to avoid?
- What evidence should we collect before changing our strategy?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected match strategy,
- chosen launch site and backup launch site,
- hopper goal and expected number of balls per scoring run,
- expected number of scoring cycles,
- autonomous goal,
- parking plan,
- penalty-avoidance focus,
- one major trade-off,
- evidence the team will collect during testing.

Reflection

Our main strategy is:

We chose this strategy because:

The main trade-off we accepted is:

The evidence we will collect is:

Chassis Physics and Space Planning

The Chassis Is the Robot's Foundation

The chassis is the base of the robot. It supports the robot's weight, holds the wheels and motors, protects the electronics and battery, and creates space for future mechanisms.

A good chassis is not just strong. It also needs to be useful. The shape of the chassis affects how the robot drives, turns, carries weight, handles contact, and makes room for attachments like a shooter, hopper, trigger, or intake.

The Starting Size Constraint

FTC robots usually need to fit inside an **18 inch x 18 inch x 18 inch** starting size. This means the robot must begin the match inside that size limit.

That does not mean every robot should use the full space. A larger robot may have more room for mechanisms, but a smaller or narrower robot may move through tight spaces, park, or share the field more easily.

For the DECODE-style challenge in this course, teams should think about:

- Can the robot fit at the start?
- Can it park or share space with other robots?
- Can game pieces move around, into, or through the chassis?
- Is there room for the battery, electronics, shooter, hopper, and trigger?
- Can the team reach important parts for repair?

Supporting Loads

The chassis carries everything else on the robot. If the frame bends, twists, or comes loose, every other system becomes less reliable.

The chassis needs to support:

- motors,
- wheels,
- battery,
- control hub and electronics,
- wires,
- shooter or scoring mechanism,
- hopper or game-piece storage,
- and any extra brackets or attachments.

Student-friendly idea:

A chassis is like a table. If the table wobbles, anything built on top of it becomes harder to use.

Staying Connected

A robot frame is made from parts connected by screws, nuts, brackets, plates, and connectors. The parts need to stay connected while the robot drives, turns, stops, bumps into objects, and carries weight.

Common chassis problems include:

- screws loosening,
- brackets bending,
- frame corners twisting,
- wheels rubbing,
- wires getting pulled,
- attachments shaking loose,
- or the battery sliding around.

Good teams check connections after testing. A robot that works once but falls apart after five minutes is not reliable yet.

Withstanding Contact

Robots may bump into game objects, field walls, or other robots. The chassis should be strong enough to handle normal contact without losing shape.

Questions to ask:

- Are important parts protected?
- Are wires away from wheels and moving parts?
- Are battery and electronics secure?
- Are corners strong enough?
- If the robot pushes a ball or artifact, does the frame twist?
- If another robot bumps us, will our mechanism stay aligned?

This does not mean the robot must be heavy or overbuilt. It means the frame should be strong enough for the job.

Motor Placement and Turning

Where the drive motors and drive wheels are placed changes how the robot feels to drive.

If the drive wheels are closer to the middle of the robot, the robot may turn more easily and may be able to rotate in place more smoothly. This can make the robot easier for a human driver to control.

If the drive wheels are farther apart or closer to the ends, the robot may feel more stable, but it may take more space to turn. It may also be harder to line up quickly.

Motor placement affects:

- turning radius,
- ability to rotate in place,
- how easy the robot is to control,
- how much space remains inside the frame,
- where the battery and electronics can go,
- how easy it is to add attachments later,
- and how protected the motors are.

There is no perfect motor placement. Teams need to decide what matters most: control, space, strength, repair access, or future mechanisms.

Driving in Multiple Directions

Most course robots will use two powered drive wheels and two support wheels, often omni wheels. This kind of robot can drive forward and backward and turn. It usually cannot slide sideways like a competition robot with mecanum wheels.

That means the chassis shape and motor placement matter. If the robot cannot move sideways, it must turn and line up carefully. A driver needs enough control to approach the goal, reach the human operator, avoid other robots, and park.

Clearance Off the Floor

Clearance means how much space there is between the bottom of the robot and the floor.

Too little clearance can cause:

- the robot to scrape,
- screws or brackets to catch on field elements,
- wheels to lose contact,
- or game pieces to jam under the frame.

Too much clearance can raise the robot's weight and make it less stable.

Good teams check whether the robot can move over the field surface without dragging or catching.

Space for Attachments

The chassis should leave room for future systems. A team may need space for:

- shooter,
- hopper,
- trigger servo,
- ball or artifact path,
- wiring,
- battery removal,
- electronics access,
- and repairs.

An open chassis may leave space for game pieces to enter or move through. A closed chassis may be stronger and easier to build. This is a trade-off.

Could Challenge: Add More Physics

If your team is ready for a challenge, you can explain the chassis using more formal physics ideas:

- **center of mass:** where the robot's weight is balanced,
- **torque:** turning force from motors or from weight placed far from the center,
- **friction:** grip between wheels and floor,

- **normal force:** how weight is shared across wheels,
- **stability:** how hard it is to tip the robot,
- **turning radius:** the space needed to complete a turn,
- **structural rigidity:** how much the frame resists bending or twisting.

You do not need these terms to build a first chassis, but they can help explain why one design works better than another.

Big Idea

The chassis is a set of engineering decisions. Shape, size, motor placement, clearance, and open space all affect how the robot works. A good chassis supports the robot, survives normal contact, leaves room for future mechanisms, and is easy enough to repair and improve.

Think About It

What should your team prioritize in the chassis: strength, open space, turning control, repair access, or room for future mechanisms?

Chassis Shape Menu

What Shape Should Our Robot Base Have?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different frame shapes and imagine how each one could support the robot's job.
- **[Red: Critical Thinking & Problem Solving]** We compare trade-offs before building.
- **[Blue: Effective Communication]** We explain our design choice in the engineering portfolio.
- **[Green: Personal Development]** We make responsible choices based on time, skill, tools, and materials.

Decision Being Made

Your team is choosing the basic shape of the robot chassis. The chassis is the base of the robot. It holds the wheels, motors, battery, electronics, and future mechanisms such as a shooter, hopper, trigger, or intake.

A chassis is not just "the bottom of the robot." Its shape affects:

- how the robot drives,
- how stable the robot is,
- how much space is available inside,
- where balls or artifacts can move,
- where motors and electronics can go,
- how easy the robot is to repair,
- and how easy it will be to add mechanisms later.

Background Sources

Use ideas from:

- chassis design discussion,
- wheel and drive system notes,
- borrowed materials expectations,
- FIRST / FTC values,
- your team's strategy ideas,
- and any robot driving observations from Day 1.

Why This Decision Matters

Teams often want to start building immediately, but chassis shape is one of the most important early robot decisions. A strong, simple frame may drive well but leave less room for game pieces. An open frame may support an intake or pushing strategy but may need extra support to stay rigid. A narrow frame may park or share space more easily but may leave less room for electronics and mechanisms.

There is no perfect shape. Your job is to choose a shape that is good enough to build, test, and improve.

How to Use This Menu

- Read the chassis shape options.
- Discuss what each shape helps and what it makes harder.
- Think about where wheels, motors, battery, electronics, and future mechanisms might go.
- Choose a first chassis shape.
- Sketch your shape from the top view.
- Record your choice, reasoning, trade-off, and evidence plan in the portfolio.

Menu Options

Option A: O-Shape / Box Frame

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The frame is closed on all four sides, like a rectangle or box. It may look like an "O" from the top.

What it helps: This is often strong, simple, and easier to square. It gives the robot a clear outside frame and can protect internal parts.

Best for: Teams that want a reliable driving base, simple construction, and clear places to mount motors, wheels, battery, and electronics.

Ease of use: Easy

Likelihood of success: High

Time to implement: Fast to moderate

Material needs: Usually needs enough channel or extrusion to make a full outside frame. It may require corner brackets, plates, or inside connectors.

Possible trade-offs: A closed front may block balls, artifacts, or intake paths. It may be harder to push game pieces through the robot. Future mechanisms may need to be mounted above or outside the frame.

Evidence we might watch: Does the frame stay square? Does it drive straight? Is there enough room inside for battery, electronics, motors, and mechanisms?

Option B: H-Shape Frame

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The frame has two side rails and a cross member in the middle, like the letter "H."

What it helps: An H-shape can leave open space at the front and back. This may help with intakes, pushing balls, or interacting with game pieces from either side.

Best for: Teams that want openings at both ends of the robot or want to leave the front and back flexible for future mechanisms.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Material needs: Needs two strong side rails and at least one cross member. The team may need brackets or plates to keep the frame rigid.

Possible trade-offs: The frame may twist more than a closed box frame. The middle cross member may block the path for a ball, hopper, battery, or electronics. The team must decide what the open spaces are for.

Evidence we might watch: Does the frame twist during turns? Can balls or artifacts move through the open spaces? Does the cross member block anything important?

Option C: A-Shape / One-End-Open Frame

Graduate Expectation Connection: [Purple: Curiosity & Creativity]

What it means: The frame is closed on one end and open on the other. It is like an H-shape where one end is closed. In this course, "A-shape" means one end is open for interaction with game pieces; it does not need to be angled.

What it helps: The open end can guide balls or artifacts, make space for an intake, or allow the robot to push game pieces toward the human operator or scoring area. The closed end can add strength and give the team a stable place for battery, electronics, or a shooter mount.

Best for: Teams that want one clear "front" of the robot for collecting, pushing, or interacting with game pieces.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Material needs: Needs side rails, a rear cross member, and possibly extra brackets or plates to reduce twisting at the open end.

Possible trade-offs: The open end may be less rigid. The team must clearly define the robot's front and protect wheels, wires, and mechanisms near the opening.

Evidence we might watch: Does the open end help with balls or artifacts? Does the frame twist? Can the team still mount wheels, motors, electronics, and battery safely?

Option D: Narrow Frame

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The robot is intentionally built narrower than the maximum allowed size.

What it helps: A narrow robot may fit through tighter spaces, share the field with other robots, park more easily, or move around crowded areas.

Best for: Teams that value maneuverability, parking, passing, or working near other robots.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Material needs: May use shorter channels or extrusions, but internal space becomes limited.

Possible trade-offs: Less room for battery, electronics, shooter, hopper, trigger, or intake. The robot may be less stable if it becomes tall or top-heavy.

Evidence we might watch: Can the robot still fit all required parts? Does it tip during turns? Does the narrow shape improve parking or movement?

Option E: Wide / Full-Size Frame

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The robot uses most of the allowed footprint.

What it helps: A wider frame gives more room for mechanisms, electronics, battery, and stable mounting. It may be easier to build and more stable.

Best for: Teams that need room for mechanisms and want a stable base.

Ease of use: Easy to medium

Likelihood of success: High

Time to implement: Moderate

Material needs: May require longer channels or extrusions. Long pieces may be limited.

Possible trade-offs: A wide robot may have trouble sharing space, parking with other robots, or moving through tight areas.

Evidence we might watch: Does the extra space help the build? Does the size create problems with parking, turning, or crowding?

Shape Comparison Table

Shape	Best Feature	Main Risk	Good For
O-shape / box	Strong and simple	May block game-piece paths	Reliable driving base
H-shape	Open front and back	May twist or block middle path	Flexible intake/pushing options
A-shape / one-end-open	One clear open end	Open end may need support	Front intake, pushing, or guiding
Narrow frame	Easier to fit in tight spaces	Less room inside	Parking, passing, shared field space
Wide / full-size frame	Stable and roomy	Harder to share space	Mechanisms and stability

Materials Note

Your team may use parts from different systems, including **goBILDA**, **REV**, and **Tetrix**. These systems may use different hole patterns, sizes, connectors, and fasteners.

You may have access to:

- U-channel from goBILDA, REV, or Tetrix,
- REV 15 mm extrusion,
- brackets,
- plates,

- inside connectors,
- bolts, nuts, and spacers.

You will not have unlimited access to every length or every part. Your chassis shape must work with the materials that are actually available.

This is part of engineering. Real teams design under constraints.

Chassis Shape Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Chassis shape				
Robot width				
Open spaces				
Cross members / support				

Sketch Requirement

Draw a top-view sketch of your chassis plan.

Your sketch should label:

- front of robot,
- back of robot,
- frame shape,
- drive wheels,
- omni wheels,
- motor locations,
- battery location,
- electronics location,
- open space for balls, artifacts, intake, shooter, hopper, or trigger.

Must / Should / Could

Level	Expectation
Must Do	Choose a chassis shape and draw a labeled top-view sketch.
Should Do	Explain why the shape fits your team's strategy and name at least one trade-off.

Level	Expectation
Could Do	Compare two chassis shapes, explain how each would affect scoring or future mechanisms, and choose using evidence or measurements.

SLD Connection

Main SLD Prompt

Should our chassis be simple and strong, or open and flexible for future mechanisms?

Supporting Questions

- What does our robot need to do first?
- Where might balls or artifacts need to move?
- Where will the battery and electronics go?
- What shape gives us the best chance of building a working robot quickly?
- What shape leaves room for future mechanisms?
- What material constraints might change our first choice?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected chassis shape,
- labeled top-view sketch,
- explanation of why the team chose the shape,
- trade-off statement,
- materials the team expects to use,
- evidence the team will watch during testing.

Reflection

Our chosen chassis shape is:

We chose this shape because:

The main trade-off we accepted is:

The evidence we will watch during testing is:

Chassis Material Menu

What Should We Build the Frame From?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different building systems and ways to solve the same design problem.
- **[Red: Critical Thinking & Problem Solving]** We compare material trade-offs before building.
- **[Green: Personal Development]** We make responsible choices based on time, skill, tools, and available materials.
- **[Orange: Cultural Understanding & Civic Engagement]** We use shared and borrowed materials responsibly.
- **[Blue: Effective Communication]** We explain our material choice in the engineering portfolio.

Decision Being Made

Your team is choosing what materials or building system to use for the robot chassis. The chassis shape matters, but the material also matters. Different systems have different strengths, fasteners, hole patterns, sizes, connectors, and tools.

Your team may not have unlimited access to every part or every length. This is normal. Real engineering teams design with the materials, money, time, and tools they actually have.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Physics and Space Planning**
- **Borrowed Equipment and Community Trust**
- materials available in the classroom,
- mentor demonstrations,
- and your team's chassis sketch.

Why This Decision Matters

The frame material affects:

- how strong the chassis is,
- how easy it is to build,
- how easy it is to add attachments,
- how much space the frame uses,
- how easy the robot is to repair,
- how well parts from different systems fit together,

- how much time the team spends on the base instead of scoring mechanisms,
- and how responsibly the team uses shared or borrowed materials.

There is no perfect material choice. Your team should choose the material that best fits your design, skill level, available parts, and time.

How to Use This Menu

- Review the material options.
- Compare strength, ease of use, time, and attachment options.
- Check what materials are actually available.
- Decide whether your team should build a new chassis, modify an existing chassis, or recycle a known chassis.
- Record the choice, trade-off, and evidence plan in the engineering portfolio.

Menu Options

Option A: goBILDA U-Channel

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: goBILDA U-channel is a rigid metal channel with many holes for attaching brackets, motors, wheels, shafts, and other parts.

What it helps: It is strong, precise, and designed for robotics. It works well for sturdy frames and reliable mounting.

Best for: Teams that want a strong chassis with many attachment points.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: goBILDA channel, compatible screws, nuts, brackets, plates, hubs, shafts, and hex tools.

Possible trade-offs: Specific lengths may not be available. Some parts may not line up cleanly with REV or Tetrix parts. Working inside channel can become hard after the robot is assembled.

Evidence we might watch: Does the frame stay rigid? Are attachments easy to add? Can we reach screws, motors, battery, and electronics?

Option B: REV U-Channel

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: REV U-channel is a metal channel used in the REV building system. It provides structure and attachment points.

What it helps: It can create strong frame members and may work well with REV brackets, plates, shafts, motors, and electronics.

Best for: Teams using mostly REV parts or needing a simple channel-based frame.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Material / tooling needs: REV channel, compatible fasteners, brackets, plates, hubs, shafts, and hex tools.

Possible trade-offs: Hole patterns and dimensions may not match goBILDA or Tetrix parts perfectly. Teams may need adapters or creative brackets.

Evidence we might watch: Can we attach wheels, motors, and mechanisms securely? Do parts line up well enough without forcing them?

Option C: REV 15 mm Extrusion

Graduate Expectation Connection: [Purple: Curiosity & Creativity]

What it means: REV 15 mm extrusion is a smaller beam-like structure that allows parts to be placed in many positions along the rail.

What it helps: Extrusion can be flexible for mounting because parts can often slide or be placed continuously instead of only at fixed holes. It is compact and useful for adjustable structures.

Best for: Teams that need flexible attachment points, small structures, supports, or adjustable mounts.

Ease of use: Medium to hard

Likelihood of success: Medium

Time to implement: Moderate to slow

Material / tooling needs: REV extrusion, brackets, corner connectors, compatible screws, nuts, and tools.

Possible trade-offs: It may be less obvious for beginners to assemble. It may require special connectors. A full chassis made only from extrusion may need careful bracing to stay square and rigid.

Evidence we might watch: Can we attach parts where we want? Does the frame stay square? Does anything slide, loosen, or twist during driving?

Option D: Tetrix Channel or Mixed Channel Parts

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: Tetrix and other channel parts may be available. Teams may combine parts from different systems when the exact preferred material is not available.

What it helps: Mixed systems can make use of available materials and may solve a parts shortage.

Best for: Teams that are flexible, willing to test fit, and able to use brackets or plates to connect parts from different systems.

Ease of use: Hard

Likelihood of success: Medium

Time to implement: Slow

Material / tooling needs: Tetrix channel, compatible fasteners, adapters, brackets, plates, and possibly parts from multiple systems.

Possible trade-offs: Different systems may not line up cleanly. Holes, screws, shafts, hubs, and spacing may not match. Teams may spend extra time solving connection problems.

Evidence we might watch: Are connections secure? Are parts being forced together? Does the frame stay aligned after driving?

Option E: Hybrid Frame

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The team intentionally combines systems. For example, the main chassis might use goBILDA channel, while smaller adjustable mounts use REV extrusion.

What it helps: A hybrid frame can use the strengths of different systems. Strong channel can support the base, while extrusion or brackets can help with adjustable mechanisms.

Best for: Teams that have a clear reason for mixing systems and enough patience to manage fit issues.

Ease of use: Hard

Likelihood of success: Medium

Time to implement: Slow

Material / tooling needs: Parts from multiple systems, adapter plates or brackets, compatible fasteners, and careful test fitting.

Possible trade-offs: Mixing systems can create alignment, spacing, and fastener problems. It may take more time than using one system.

Evidence we might watch: Does each system solve a specific problem? Are mixed connections strong and repairable?

Option F: Recycle or Modify a Known Chassis

Graduate Expectation Connection: [Green: Personal Development], [Orange: Cultural Understanding & Civic Engagement], and [Red: Critical Thinking & Problem Solving]

What it means: Instead of building a brand-new chassis from scratch, the team uses an existing chassis, known base, or partially built robot frame and modifies it for the challenge.

What it helps: This can save time, reduce risk, and give the team more time for driving, programming, shooter design, hopper/trigger work, testing, and portfolio development.

Best for: Teams that want a reliable starting point, have limited build time, or need to focus on scoring mechanisms and controls.

Ease of use: Easy to medium

Likelihood of success: High if the chassis is in good condition

Time to implement: Fast

Material / tooling needs: Existing chassis, inspection checklist, repair tools, possible replacement screws, wheels, hubs, motors, or brackets.

Possible trade-offs: The team may have less ownership over the original design. The existing chassis may not perfectly fit the team's strategy. Old choices may limit where attachments, battery, electronics, or game-piece paths can go.

Evidence we might watch: Does the existing chassis drive reliably? What design limits does it create? How much time did it save? What did the team modify?

Comparison Table

Option	Best Feature	Main Risk	Good For
goBILDA U-channel	Strong and precise	Specific lengths/parts may be limited	Strong chassis base
REV U-channel	Works well with REV ecosystem	May not align with other systems	REV-based builds
REV 15 mm extrusion	Flexible placement	Needs careful connectors/bracing	Adjustable mounts and compact structures
Tetrix / mixed channel	Uses available parts	Fit and fastener problems	Teams solving material shortages
Hybrid frame	Uses strengths of multiple systems	More complex connections	Teams with a clear mixed-system plan
Recycled known chassis	Saves time and reduces risk	Less original design control	Teams prioritizing mechanisms, code, and testing

Materials Availability Check

Before finalizing your material choice, check:

- Do we have enough pieces in the lengths we need?
- Do we have the correct fasteners?
- Do we have brackets or connectors that fit?
- Do we have the correct hubs, shafts, spacers, and wheels?
- Can we attach motors securely?
- Can we reach screws later for repair?
- Are we using borrowed materials responsibly?
- Do we need to request a limited part?

Material Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Main frame material				
Connector/bracket plan				
Mixed systems?				

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Recycled chassis?				

Must / Should / Could

Level	Expectation
Must Do	Choose a chassis material or existing chassis and explain why it is good enough to start building.
Should Do	Explain the main trade-off of the material choice and check whether the needed parts are available.
Could Do	Compare two material systems or compare new-build vs. recycled chassis using time, reliability, repair access, and future mechanism space.

SLD Connection

Main SLD Prompt

Should our team build a new chassis from preferred materials, adapt to the materials available, or recycle a known chassis to save time?

Supporting Questions

- What material gives us the best chance of a reliable driving base?
- What material gives us the most flexibility for future attachments?
- What material is actually available?
- How much time should we spend on the chassis before moving to mechanisms?
- When is recycling a known chassis a smart engineering decision?
- What do we lose if we do not build the chassis from scratch?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected chassis material or existing chassis,
- reason for the choice,
- material trade-off statement,
- parts availability notes,
- sketch or photo of planned frame,
- and evidence the team will watch during testing.

Reflection

Our chassis material choice is:

We chose this because:

The main trade-off we accepted is:

One material constraint we need to manage is:

Wheel Selection Menu

How Should Our Robot Touch the Field?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different wheel types and imagine how they affect robot movement.
- **[Red: Critical Thinking & Problem Solving]** We compare traction, turning, speed, control, and reliability.
- **[Green: Personal Development]** We make responsible choices based on available materials and team skill.
- **[Blue: Effective Communication]** We explain wheel choices using sketches, test data, and observations.

Decision Being Made

Your team is choosing the wheels for the chassis. Wheels affect how the robot drives, turns, accelerates, pushes, stops, and handles contact with the field or game objects.

Teams may have access to wheels from different systems, including Tetrix, REV, goBILDA, and others. Wheel sizes, tread patterns, rubber density, hubs, axles, and connectors may all be different.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Material Menu**
- **Chassis Physics and Space Planning**
- robot driving observations,
- available wheel bins,
- mentor demonstrations,
- and your team's chassis sketch.

Why This Decision Matters

Wheels are not just "the things that roll." They determine how the robot interacts with the floor.

Wheel choice affects:

- traction,
- turning,
- speed,
- stability,
- pushing ability,
- driver control,
- autonomous consistency,

- clearance,
- and how easily the wheels attach to motors, hubs, shafts, and the chassis.

A wheel that looks good may not fit the shaft. A wheel that grips well may make turning harder. A wheel that turns easily may not push well. Your team needs to choose based on the job the robot needs to do.

How to Use This Menu

- Review the wheel options.
- Check which wheels are actually available.
- Check whether the wheels fit your hubs, shafts, axles, and chassis.
- Decide which wheels are powered and which wheels support the robot.
- Sketch the wheel layout.
- Record your choice, reasoning, trade-off, and evidence plan in the portfolio.

Wheel Design Factors

Wheel Size

Larger wheels can move the robot farther with each rotation, which can make the robot faster. They can also increase clearance off the floor. Smaller wheels may be easier to fit and may give more control.

Larger Wheels	Smaller Wheels
More distance per rotation	More compact
More floor clearance	Often easier to fit
May increase speed	May feel easier to control
May need more torque	May reduce clearance

Tread and Rubber

Soft rubber and grippy tread can help the robot push and accelerate. Harder wheels or smoother wheels may slip more but can sometimes turn more easily.

More Grip	Less Grip
Better pushing and acceleration	Easier turning in some layouts
More control when driving straight	Less strain during turns
May resist sideways movement	May slip under load
Can make turning harder	May reduce pushing power

Fit and Connection

Before choosing a wheel, check:

- Does the wheel fit the hub?

- Does the hub fit the shaft or axle?
- Is the shaft round or hex?
- Is the set screw tight?
- Is the wheel supported?
- Does the wheel rub the frame?
- Do we have spacers to stop side-to-side sliding?
- Does the wheel match the motor placement and chassis shape?

Menu Options

Option A: Standard Traction Wheels

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: Traction wheels are regular wheels designed to grip the floor.

What it helps: They provide pushing power, acceleration, and strong forward/backward movement.

Best for: Drive wheels powered by motors.

Ease of use: Easy to medium

Likelihood of success: High if they fit the hub and shaft

Time to implement: Fast to moderate

Material / tooling needs: Wheels, hubs, shafts/axles, set screws, spacers, bearings or supports, and hex tools.

Possible trade-offs: Too much grip can make turning harder, especially if all four wheels are traction wheels.

Evidence we might watch: Does the robot drive straight? Does it turn smoothly? Do the wheels slip or scrub during turns?

Option B: Omni Wheels

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: Omni wheels have small rollers around the edge. They roll forward like normal wheels, but can slide sideways more easily.

What it helps: Omni wheels reduce sideways resistance during turns. A common beginner layout uses two powered traction wheels and two omni support wheels.

Best for: Helping a two-motor robot turn more smoothly.

Ease of use: Medium

Likelihood of success: High if installed correctly

Time to implement: Moderate

Material / tooling needs: Omni wheels, compatible hubs, shafts/axles, spacers, and supports.

Possible trade-offs: Omni wheels usually provide less sideways resistance, so the robot may be easier to push sideways. They may provide less traction than regular wheels.

Evidence we might watch: Does the robot turn more easily? Does it drift sideways? Does it still drive straight enough for testing?

Option C: Four Traction Wheels

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: All four wheels are regular traction wheels.

What it helps: This can give the robot strong grip and pushing power.

Best for: Robots that need to push or hold position and do not need to turn very tightly.

Ease of use: Easy

Likelihood of success: Medium

Time to implement: Fast

Material / tooling needs: Four matching traction wheels, hubs, axles/shafts, spacers, and supports.

Possible trade-offs: The robot may fight itself while turning because all four wheels resist sideways sliding. This is called wheel scrub. It can make turning jerky, slow, or hard on motors.

Evidence we might watch: Does the robot turn smoothly? Do motors strain during turns? Does the robot jerk or skid?

Option D: Two Traction Wheels + Two Omni Wheels

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: Two powered traction wheels move the robot. Two omni wheels support the robot and help it turn.

What it helps: This is a strong beginner layout because it balances powered movement with easier turning.

Best for: Two-motor course robots that need simple, reliable driving.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: Two traction wheels, two omni wheels, compatible hubs/shafts/spacers, and supports.

Possible trade-offs: Less pushing power than four traction wheels. The robot may slide more easily from the omni-wheel side.

Evidence we might watch: Does the robot turn predictably? Does it drive straight? Can the driver control it easily?

Option E: Mecanum Wheels

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: Mecanum wheels have angled rollers that allow a robot to drive forward/backward, turn, and move sideways when used in a four-motor drivetrain.

What it helps: Mecanum drive can make it easier to line up with goals, move around field objects, and strafe sideways.

Best for: Advanced teams with enough motors, wheels, time, and programming support.

Ease of use: Hard

Likelihood of success: Risky for a short course unless the base already exists

Time to implement: Slow

Material / tooling needs: Four mecanum wheels, four drive motors, correct wheel orientation, motor controllers, more complex code, tuning, and driver practice.

Possible trade-offs: More expensive, more complex, more programming, more wiring, and more debugging. It may take time away from shooter, hopper, trigger, portfolio, and testing.

Evidence we might watch: Can the robot strafe reliably? Do all wheels spin in the correct direction? Can the driver control it under pressure?

Option F: Reused Wheels From an Existing Chassis

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: The team uses wheels already mounted on a known chassis or salvages wheels from an existing robot base.

What it helps: This can save time and reduce fit problems because the wheels, hubs, shafts, and supports may already work together.

Best for: Teams that want to prioritize robot reliability, programming, mechanisms, or testing time.

Ease of use: Easy to medium

Likelihood of success: High if the wheels are in good condition

Time to implement: Fast

Material / tooling needs: Existing chassis or wheel assemblies, inspection checklist, hex tools, replacement screws, spacers, or hubs if needed.

Possible trade-offs: The wheel layout may limit chassis shape, motor placement, ground clearance, or future mechanisms.

Evidence we might watch: Do the wheels spin freely? Are set screws tight? Does the robot drive straight and turn reliably?

Wheel Layout Questions

Before finalizing the wheel choice, answer:

- Which wheels are powered by motors?
- Which wheels are support wheels?
- Where are the omni wheels, if used?
- Do the wheels fit the hubs and shafts?
- Are the set screws tight?
- Is the wheel supported on the axle?
- Does the wheel rub the frame?

- Is there enough clearance off the floor?
- Will the robot turn smoothly?
- Will the driver be able to control the robot?

Wheel Assembly Checklist

Check	Yes / No
Wheel fits the hub	
Hub fits the shaft or axle	
Set screw is tight	
Wheel spins freely	
Wheel does not rub the frame	
Spacers stop side-to-side sliding	
Axle or shaft is supported	
Left and right drive wheels are aligned	
Omni wheels roll in the correct direction	
Robot has enough floor clearance	

Comparison Table

Option	Best Feature	Main Risk	Good For
Standard traction wheels	Grip and pushing	Turning may be harder	Powered drive wheels
Omni wheels	Easier turning	Less sideways resistance	Support wheels on simple drivetrains
Four traction wheels	Strong grip	Wheel scrub during turns	Pushing or holding position
Two traction + two omni	Balanced beginner drive	Less pushing than four traction	Reliable two-motor robot
Mecanum wheels	Sideways movement	Complex, expensive, needs four motors	Advanced competition-style driving
Reused wheel assemblies	Saves time	Limits design choices	Reliable fast start

Wheel Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Drive wheels				

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Support wheels				
Wheel size				
Wheel material/tread				
Wheel layout				

Must / Should / Could

Level	Expectation
Must Do	Choose wheels that fit the chassis and identify which wheels are powered.
Should Do	Explain how the wheel choice affects turning, traction, control, and reliability.
Could Do	Compare two wheel layouts using a drive test, turning test, or driver feedback.

SLD Connection

Main SLD Prompt

Should our robot prioritize traction, easy turning, speed, or driver control when choosing wheels?

Supporting Questions

- What wheel choice gives us the best chance of driving reliably?
- When is more grip helpful?
- When can too much grip make turning harder?
- Should we use omni wheels to make turning easier?
- Is mecanum worth the extra complexity in a short course?
- Should we reuse a known wheel setup to save time?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected wheel types,
- powered wheel locations,
- support wheel locations,
- wheel layout sketch,
- wheel assembly checklist,

- explanation of the main trade-off,
- and evidence the team will watch during testing.

Reflection

Our wheel choice is:

We chose this because:

The main trade-off we accepted is:

The evidence we will watch during testing is:

Motor Placement Menu

Where Should the Drive Motors Go?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different ways to fit motors into the chassis.
- **[Red: Critical Thinking & Problem Solving]** We compare turning, space, attachment access, and repair trade-offs.
- **[Green: Personal Development]** We make responsible choices based on available parts, time, and team skill.
- **[Blue: Effective Communication]** We explain motor placement using sketches, evidence, and trade-offs.

Decision Being Made

Your team is deciding where to place the drive motors. Motor placement affects how the robot turns, how easy it is to control, how much open space remains in the chassis, how easy it is to attach future mechanisms, and how easy the robot is to repair.

The drive motors are not just power sources. Their location shapes the whole robot layout.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Material Menu**
- **Wheel Selection Menu**
- **Chassis Physics and Space Planning**
- robot driving observations,
- available motors, hubs, gears, shafts, and brackets,
- and your team's chassis sketch.

Why This Decision Matters

Motor placement is one of the hidden decisions that affects everything else.

Motors in the center may make the robot easier to turn, but they take up valuable middle space. Motors near the ends may leave the center open, but the robot may need more room to turn. Motors inside channels may look clean and protected, but they can make it harder to add attachments later.

Your team needs to decide what matters most: turning, space, protection, repair access, future mechanisms, or speed of building.

How to Use This Menu

- Review the motor placement options.
- Check your chassis shape and wheel layout.
- Think about where the battery, electronics, shooter, hopper, and trigger might go.

- Choose a first motor placement plan.
- Sketch the motors on your top-view chassis drawing.
- Record your choice, trade-off, and evidence plan in the portfolio.

Menu Options

Option A: Motors Near the Center

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The drive motors and powered wheels are placed closer to the middle of the robot.

What it helps: This can make the robot easier to turn and may help it rotate in place more smoothly. It can make the robot feel more controllable for a human driver.

Best for: Teams that want easier turning, predictable control, and a robot that can line up without needing a huge turning space.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: Motor mounts, compatible wheels, hubs, shafts, brackets, spacers, and space in the middle of the chassis.

Possible trade-offs: Motors in the center occupy valuable middle space. This may interfere with a ball path, hopper, shooter mount, battery, electronics, or future intake.

Evidence we might watch: Does the robot turn smoothly? Can it rotate in place? Is there still enough room for the battery, electronics, and future mechanisms?

Option B: Motors Near the Ends

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The motors and powered wheels are placed closer to the front and/or back ends of the robot.

What it helps: This can leave the center of the robot more open for game pieces, electronics, battery, hopper, or mechanisms.

Best for: Teams that need the middle of the robot open for a ball path, hopper, shooter, or intake.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Material / tooling needs: Motor mounts, wheels, hubs, shafts, brackets, and wiring that can reach the motors safely.

Possible trade-offs: The robot may be harder to turn. It may need more space to rotate, and the driver may need more practice to control it.

Evidence we might watch: Does the robot need too much room to turn? Does it line up accurately? Does the open middle space help with future mechanisms?

Option C: Motors Along the Side Rails

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: Motors are mounted along the left and right sides of the chassis, usually near the wheels but not necessarily inside the channel.

What it helps: This can make motors easier to see, wire, repair, and adjust. It may keep the center more open while still supporting a simple drive layout.

Best for: Teams that want repair access and a clear, understandable drivetrain.

Ease of use: Easy to medium

Likelihood of success: High

Time to implement: Fast to moderate

Material / tooling needs: Side mounting brackets, motor mounts, wheels, hubs, spacers, and wire management.

Possible trade-offs: Motors may stick out or be more exposed to contact. They may reduce space for side attachments.

Evidence we might watch: Can we reach the motors and wires? Are the motors protected enough? Do they block future attachments?

Option D: Motors Inside the Channels

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The motors are mounted inside U-channel or tucked tightly into the frame. This may use compact motor placement, bevel gears, or special brackets.

What it helps: This can protect motors and create a clean, compact drivetrain. It may keep the outside of the robot clear.

Best for: Teams that want a compact design and have enough time, tools, and support to build carefully.

Ease of use: Hard

Likelihood of success: Medium to risky

Time to implement: Slow

Material / tooling needs: Compatible channel, motor mounts, gears or shafts if needed, careful alignment, hex tools, and possibly mentor support.

Possible trade-offs: Once motors are inside the channel, it can be much harder to attach new parts to that channel. Repairs may also be harder. If the motor or gear alignment is wrong, debugging can take a long time.

Evidence we might watch: Can we still attach parts to the channel? Can we reach screws and wires? Does the drivetrain run smoothly without binding?

Option E: Motors Protected but Accessible

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The team places motors where they are partly protected by the frame but still easy enough to reach for repair, wiring, and inspection.

What it helps: This balances durability with repair access. It avoids hiding the motors so deeply that they become difficult to fix.

Best for: Teams that want a practical, reliable course robot.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: Motor mounts, brackets, protective frame members, enough space for tools and wires.

Possible trade-offs: The robot may not look as compact as a hidden-motor design, and motors may still take up useful space.

Evidence we might watch: Can the team fix a loose motor or wire quickly? Are motors protected during normal driving and contact?

Option F: Keep Motor Placement From a Recycled Chassis

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: If the team is using a known or recycled chassis, it keeps the existing motor placement unless there is a clear reason to change it.

What it helps: This saves time and reduces the risk of rebuilding the drivetrain incorrectly.

Best for: Teams that want to spend more time on programming, driving, shooter design, hopper, trigger, testing, and portfolio work.

Ease of use: Easy

Likelihood of success: High if the drivetrain already works

Time to implement: Fast

Material / tooling needs: Existing chassis, inspection checklist, replacement screws or wires if needed.

Possible trade-offs: The existing motor placement may limit chassis shape, open space, turning behavior, or future mechanisms.

Evidence we might watch: Does the recycled drivetrain drive reliably? What space does it leave or block? Would changing it cost more time than it saves?

Comparison Table

Option	Best Feature	Main Risk	Good For
Motors near center	Easier turning and rotation	Uses middle space	Driver control and tight turns
Motors near ends	Opens middle space	Harder turning	Game-piece paths and mechanisms
Motors along side rails	Easy access and clear layout	More exposed	Repairable simple drivetrains
Motors inside channels	Compact and protected	Hard to attach/repair later	Advanced compact builds
Protected but accessible	Balanced practical design	Still uses space	Reliable course robot

Option	Best Feature	Main Risk	Good For
Keep recycled placement	Saves time	Limits design choices	Fast reliable start

Motor Placement Questions

Before finalizing motor placement, answer:

- Where are the powered wheels?
- Can the robot turn in the space available?
- Can the driver control the robot easily?
- Does motor placement block the center of the robot?
- Is there room for the battery and electronics?
- Is there room for the shooter, hopper, trigger, or intake?
- Can we reach motor screws and wires later?
- Are motors protected from normal contact?
- Are wires safe from wheels and moving parts?
- Does this placement fit our chassis shape and materials?

Motor Placement Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Motor placement				
Powered wheel location				
Space affected				
Repair access				

Must / Should / Could

Level	Expectation
Must Do	Choose where the drive motors will go and show them on a labeled chassis sketch.
Should Do	Explain how motor placement affects turning, driver control, space, and repair access.
Could Do	Compare two motor placements using turning radius, driver feedback, or available space measurements.

SLD Connection

Main SLD Prompt

Should our drive motors be placed for easier turning, more open space, better protection, or easier repair?

Supporting Questions

- Is turning control more important than open space?
- What future mechanisms might need the middle of the chassis?
- How much repair access do we need?
- Should motors be protected even if they become harder to reach?
- What motor placement best fits our chassis shape and wheel choice?
- When is a clean compact design worth the extra build difficulty?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected motor placement,
- labeled chassis sketch showing motors and powered wheels,
- explanation of why the team chose this placement,
- trade-off statement,
- repair/access notes,
- and evidence the team will watch during testing.

Reflection

Our motor placement choice is:

We chose this because:

The main trade-off we accepted is:

The evidence we will watch during testing is:

Electronics and Battery Placement Menu

Where Should the Hub, Battery, Switch, and Wires Go?

Graduate Expectation Tags

- **[Red: Critical Thinking & Problem Solving]** We compare balance, access, wiring, repair, and safety trade-offs.
- **[Green: Personal Development]** We make responsible choices that protect electronics and support safe testing.
- **[Blue: Effective Communication]** We explain our layout clearly in sketches, tables, and portfolio notes.
- **[Orange: Cultural Understanding & Civic Engagement]** We protect borrowed electronics and shared equipment.

Decision Being Made

Your team is deciding where to place the robot's electronics and battery. These parts are essential. If the hub, battery, switch, or wires are hard to reach or poorly mounted, the robot will be harder to test, repair, and improve.

Important parts include:

- Control Hub or Expansion Hub,
- battery,
- power switch,
- motor wires,
- servo wires,
- sensor wires if used,
- and USB or communication cables if needed.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Material Menu**
- **Motor Placement Menu**
- **Chassis Physics and Space Planning**
- **Robotics Safety Norms**
- **Borrowed Equipment and Community Trust**
- your team's chassis sketch.

Why This Decision Matters

Electronics placement affects:

- whether wires can reach motors and servos,
- whether the battery can be changed quickly,
- whether the power switch is easy to reach,
- whether the robot is balanced,
- whether the hub lights and ports are visible,
- whether damaged parts can be replaced,
- whether future mechanisms can be added,
- and whether electronics are protected from contact.

Good electronics layout is not just about making the robot look clean. It is about making the robot reliable, repairable, and safe.

Required Layout Rule

The **power switch must be between the battery and the hub** and must be easy to reach. A teammate, mentor, or inspector should be able to find and use the switch quickly.

The battery should also be easy enough to remove and replace, because fresh batteries are needed often.

Menu Options

Option A: Hub Mounted Flat on the Floor of the Chassis

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: The hub is mounted flat on a low plate or frame surface inside the robot.

What it helps: This can keep the hub low and protected. It may lower the robot's center of mass and make the robot more stable.

Best for: Teams that want a low, protected electronics layout.

Ease of use: Easy to medium

Likelihood of success: Medium to high

Time to implement: Moderate

Possible trade-offs: Top ports may be easy to see, but bottom or side connectors may be harder to reach depending on the hub orientation. If mechanisms are built above it, access may become difficult.

Evidence we might watch: Can we plug and unplug wires easily? Can we see hub lights? Can we replace or inspect wires after mechanisms are added?

Option B: Hub Mounted Vertically on a Side Wall

Graduate Expectation Connection: [Blue: Effective Communication] and [Red: Critical Thinking & Problem Solving]

What it means: The hub is mounted upright on the side of the chassis or on a vertical plate.

What it helps: This can make ports, labels, wires, and lights easier to see. It may also keep the floor of the robot open for game pieces or mechanisms.

Best for: Teams that expect to add motors, servos, sensors, or wires during the course.

Ease of use: Medium

Likelihood of success: High if mounted securely

Time to implement: Moderate

Possible trade-offs: The hub may be more exposed to contact. It may raise some weight higher in the robot. It needs a strong mount so it does not shake loose.

Evidence we might watch: Can we reach the ports? Are wires organized? Is the hub protected enough during normal driving?

Option C: Hub Mounted High for Easy Access

Graduate Expectation Connection: [Green: Personal Development]

What it means: The hub is mounted where students can easily see and reach ports without removing other parts.

What it helps: This makes debugging, adding parts, checking wires, and reading lights easier.

Best for: Teams that expect frequent changes to wiring and code configuration.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Possible trade-offs: Mounting electronics higher can raise the center of mass. It may also expose the hub to contact or game pieces.

Evidence we might watch: Can we troubleshoot quickly? Does the robot remain stable during turns and contact?

Option D: Hub Hidden and Protected Inside the Frame

Graduate Expectation Connection: [Orange: Cultural Understanding & Civic Engagement]

What it means: The hub is placed inside the robot where frame pieces or plates protect it from contact.

What it helps: This can protect expensive borrowed electronics from bumps, game pieces, and robot contact.

Best for: Teams that prioritize protecting electronics and have already planned wire access carefully.

Ease of use: Hard

Likelihood of success: Medium

Time to implement: Slow

Possible trade-offs: Hidden electronics can become hard to repair, inspect, or rewire. If a wire comes loose, the team may need to remove other parts to fix it.

Evidence we might watch: Can we still reach every important port? Can we replace damaged parts? Can we see the status lights?

Option E: Battery Low and Centered

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The battery is mounted low in the robot, near the center of the chassis.

What it helps: The battery is heavy. Placing it low and near the center can improve balance and reduce tipping.

Best for: Teams that want stable driving and predictable turning.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Possible trade-offs: The battery may be harder to remove if it is buried inside the frame. The team must still be able to replace it quickly.

Evidence we might watch: Does the robot stay stable during turns? How long does it take to change the battery?

Option F: Battery Easy to Remove

Graduate Expectation Connection: [Green: Personal Development]

What it means: The battery is placed where students can quickly disconnect, remove, replace, and secure it.

What it helps: Battery changes happen often. Easy access saves time and reduces stress before testing or matches.

Best for: Teams that expect frequent testing and driving practice.

Ease of use: Easy to medium

Likelihood of success: High

Time to implement: Fast to moderate

Possible trade-offs: The most accessible battery location may not be the best balance location. It may also take up space needed for mechanisms.

Evidence we might watch: Can we replace the battery quickly? Is the battery secure? Does it shift during driving?

Option G: Switch on the Outside Edge

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: The power switch is mounted on an outside edge of the robot where it is easy to see and reach.

What it helps: This supports safe testing because the robot can be turned off quickly.

Best for: All teams. The switch should be easy to reach.

Ease of use: Easy

Likelihood of success: High

Time to implement: Fast

Possible trade-offs: The switch must be protected enough that it does not get hit accidentally, but it cannot be hidden.

Evidence we might watch: Can someone find and turn off the robot quickly? Is the switch protected from accidental bumps?

Option H: Electronics on One Side, Mechanisms on the Other

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The team organizes the robot so electronics and wires are mostly on one side, while mechanisms such as shooter, hopper, or trigger are on another side or area.

What it helps: This can make wiring cleaner and reduce conflicts between moving parts and electronics.

Best for: Teams that want organized wiring and clearer repair zones.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Possible trade-offs: Putting too much weight on one side can affect balance. Wires may need to travel farther to reach motors or servos.

Evidence we might watch: Does the robot pull or tip to one side? Are wires clear of moving parts? Can mechanisms move without hitting electronics?

Comparison Table

Option	Best Feature	Main Risk	Good For
Hub flat on floor	Low and protected	Harder access if covered	Stable compact layouts
Hub vertical on side	Easier port access	More exposed	Frequent wiring changes
Hub high for access	Easy troubleshooting	Higher center of mass	Debugging and additions
Hub hidden/protected	Protects electronics	Harder repair and inspection	Carefully planned layouts
Battery low/centered	Better balance	Harder battery changes	Stability
Battery easy to remove	Fast replacement	May affect balance/space	Frequent testing
Switch outside edge	Fast shutoff	Must avoid accidental hits	Safety and inspection
Electronics one side	Organized wiring	Side-heavy risk	Clean layout and repair zones

Layout Questions

Before finalizing electronics and battery placement, answer:

- Can we reach every motor, servo, and sensor port?
- Can we see hub lights?
- Can we replace a damaged wire?
- Can we add another motor or servo later?
- Can we remove and replace the battery quickly?

- Is the battery secure?
- Is the battery low enough to help balance?
- Is the power switch between the battery and hub?
- Is the power switch easy to reach?
- Are wires protected from wheels, gears, flywheels, and moving mechanisms?
- Does the layout leave room for shooter, hopper, trigger, and ball/artifact movement?

Electronics and Battery Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Hub orientation				
Hub location				
Battery location				
Switch location				
Wiring path				

Must / Should / Could

Level	Expectation
Must Do	Choose locations for the hub, battery, and switch and show them on the chassis sketch.
Should Do	Explain how the layout affects access, balance, wiring, safety, and future mechanisms.
Could Do	Compare two electronics layouts using battery-change time, wiring access, balance, or repair access.

SLD Connection

Main SLD Prompt

Should our electronics layout prioritize protection, easy access, balance, or open space for mechanisms?

Supporting Questions

- Is it better for the hub to be flat or vertical?
- Where can the battery go so the robot stays balanced but the battery is still easy to replace?
- How can the switch be easy to reach but protected from accidental hits?
- What parts will we need to add later?

- What happens if a wire comes loose during testing?
- How can we protect borrowed electronics while still making repairs possible?

Portfolio Deliverable

Add the following to your engineering portfolio:

- hub orientation and location,
- battery location,
- switch location,
- wiring path notes,
- labeled chassis sketch,
- explanation of the main trade-off,
- and evidence the team will watch during testing.

Reflection

Our hub placement choice is:

Our battery placement choice is:

Our switch placement choice is:

The main trade-off we accepted is:

The evidence we will watch during testing is:

Human-Centered Controller Mapping Background

How Should the Controller Match the Driver's Intuition?

Robot control is not only a programming problem. It is also a human design problem.

When drivers are calm, they can think carefully about what each button does. When drivers are stressed, rushed, defended, or trying to score before time runs out, they rely more on habit, muscle memory, and previous game experience.

That means a good controller map should feel predictable. The best control scheme is not always the one with the most features. It is the one the driver can use correctly under pressure.

The Big Idea

A controller map is a design decision. Teams should choose controls that match:

- what the robot needs to do,
- what the driver already understands,
- what actions are most urgent,
- what mistakes would be costly,
- and what the team can practice enough to make automatic.

If the driver has to stop and remember the controls during a match, the map is too complicated or not practiced enough.

Human Intuition and Stress

Under pressure, people often fall back on familiar patterns.

Examples:

- If a student has played console games, they may expect the left stick to move and the right stick to turn or aim.
- If a student has used D-pad controls in simple games, they may expect each direction to mean a clear fixed movement.
- If a student has driven remote-control cars, they may expect one stick or trigger to control forward/backward and another control to steer.
- If a student has practiced the same loading and shooting routine many times, they may do it smoothly without needing much verbal coaching.

This is why practice matters. A confusing control map can become manageable with training, but a simple map becomes powerful when the driver has practiced it repeatedly.

What Is the Logitech F310?

The Logitech F310 is a wired USB gamepad commonly used with FTC Driver Stations. It is designed mainly as a PC game controller.

Important features:

Feature	Why It Matters
USB wired connection	Simple and reliable for Driver Station use.
XInput mode	Makes the controller behave like a modern Windows/Xbox-style gamepad.
DirectInput mode	Older PC input mode; useful in some computer games, but usually not the first choice for FTC.
Two analog sticks	Useful for smooth driving, turning, aiming, or mechanism control.
D-pad	Useful for simple fixed-direction driving or step-by-step adjustments.
A/B/X/Y buttons	Useful for discrete actions such as shoot, open trigger, reset, slow mode, or mode change.
Bumpers and triggers	Useful for actions the driver may need while keeping thumbs on the sticks.

For FTC use, the switch on the bottom of the F310 is usually set to **X** for XInput mode. In many FTC setup examples, the driver assigns the controller by pressing **Start + A** for gamepad 1 or **Start + B** for gamepad 2.

How the F310 Relates to Xbox, PlayStation, and PC Controllers

The F310 is not an Xbox or PlayStation controller, but it borrows familiar ideas from both console-style and PC gamepads.

Controller Family	Connection to the F310
PC gamepads	The F310 is officially a PC-style USB gamepad and supports common PC input standards.
Xbox-style controllers	In XInput mode, the F310 uses Xbox-style button names: A, B, X, Y, bumpers, triggers, sticks, Start, and Back. FTC code also commonly uses these names.
PlayStation-style controllers	The physical idea is familiar: two sticks, a D-pad, four face buttons, shoulder controls, and center buttons. The labels are different: PlayStation uses symbols such as triangle, circle, cross, and square instead of A/B/X/Y.
Nintendo-style controllers	The F310 also has four face buttons and a D-pad, but button positions and labels may not match what a Nintendo player expects.

This matters because students may bring different controller habits. A driver who grew up with PlayStation may understand the layout but not the A/B/X/Y names automatically. A driver used to Xbox-style games may recognize the names more quickly.

FTC Code Names

In FTC programs, controller inputs are usually read through names like:

Driver Action	Common FTC Code Name
Left stick forward/back	<code>gamepad1.left_stick_y</code>

Driver Action	Common FTC Code Name
Left stick left/right	<code>gamepad1.left_stick_x</code>
Right stick left/right	<code>gamepad1.right_stick_x</code>
D-pad directions	<code>gamepad1.dpad_up</code> , <code>gamepad1.dpad_down</code> , <code>gamepad1.dpad_left</code> , <code>gamepad1.dpad_right</code>
Face buttons	<code>gamepad1.a</code> , <code>gamepad1.b</code> , <code>gamepad1.x</code> , <code>gamepad1.y</code>
Bumpers	<code>gamepad1.left_bumper</code> , <code>gamepad1.right_bumper</code>
Triggers	<code>gamepad1.left_trigger</code> , <code>gamepad1.right_trigger</code>

This means the code should be written in a way the team can explain. For example:

- A might mean shoot.
- B might mean stop shooter.
- Right bumper might open the trigger.
- Left bumper might close the trigger.
- Right trigger might activate slow mode.

The exact map is a team decision.

D-Pad Control

D-pad control uses the cross-shaped direction pad.

What It Helps	What It Makes Harder
Simple directions	Smooth curves
Beginner predictability	Fine speed control
Testing one movement at a time	Natural driving feel
Repeatable short moves	Fast aiming adjustments

D-pad control can be good for first tests because it is clear: up means forward, down means backward, left means turn or slide left, and right means turn or slide right. It can also feel less natural when the driver needs smooth aiming or tiny corrections.

Joystick Control

Joystick control uses one or both analog sticks.

What It Helps	What It Makes Harder
Smooth driving	Beginner overcorrection
Variable speed	Consistent straight-line movement
Natural video-game feel	Debugging if directions are reversed
Fast aiming corrections	Avoiding sudden full-power movement

Joystick control often feels more like modern console or PC games. It can be powerful, but it may need dead zones, speed scaling, or a precision mode so the robot does not jump or turn too sharply.

Speed and Precision Modes

A robot that is easy to drive at full speed may still be hard to aim. Teams can program control modes to help.

Mode	What It Does	When It Helps
Normal mode	Full driver control range	Moving across the field
Slow mode	Reduces motor power	Aiming, loading, parking, avoiding penalties
Precision turn	Reduces turning speed	Lining up with a launch site
Dead zone	Ignores tiny stick movements	Prevents drift when the stick is near center

Speed modes should be easy to remember. If slow mode is important, put it on a control the driver can reach without looking down.

Mapping Mechanisms

Driving controls should be the easiest to use because the robot is moving while the driver is thinking. Mechanism controls should also match the team's routine.

For a human-loaded ball launcher, the sequence might be:

- Drive to loading position.
- Human operator loads up to three balls.
- Driver moves to launch site.
- Driver aims.
- Driver spins up flywheel.
- Driver opens trigger to release one ball or a controlled set of balls.
- Driver repeats or returns to base.

The controller map should support this sequence.

Robot Action	Human-Centered Mapping Question
Drive	Can the driver move without looking down?
Slow/precision mode	Can the driver aim without accidentally driving too fast?
Spin up flywheel	Is the control hard to press by accident?
Open trigger	Is the control easy to press at the exact moment needed?
Stop shooter	Is there a clear safe stop action?
Reset or unjam	Is this protected from accidental use?

Safety and Error Prevention

Some controls should be easy. Some controls should be deliberate.

Control Type	Good Design Choice
Emergency stop or safe stop	Easy to find and easy to explain.
Driving	Natural and practiced.
Precision mode	Easy to hold or toggle while aiming.
Shooter spin-up	Deliberate enough to avoid accidental activation.
Trigger release	Clear and consistent.
Reset or reverse	Protected from accidental presses.

A good question is: "What is the worst mistake the driver could make with this button map?" Then design the controls to make that mistake less likely.

Training Matters

A good controller map is only good if the driver practices it.

Teams should test:

- Can the driver explain every control without looking?
- Can the driver complete a loading, driving, aiming, and shooting cycle three times?
- Does the driver accidentally press the wrong button?
- Does the driver overcorrect while aiming?
- Does the human operator understand the driver's signals?
- Can another teammate use the same map if the main driver is absent?

Main Takeaway

Controller mapping is a human-centered design decision. The code should match the driver, the robot, the game strategy, and the team's practice time.

The best explanation sounds like this:

We chose this controller map because it matched our driver's intuition, reduced mistakes under pressure, and helped us complete our scoring routine reliably.

Source Notes

This page is based on FTC controller programming conventions and Logitech F310 documentation. Logitech describes the F310 as a PC gamepad with XInput and DirectInput modes. FTC setup examples commonly use the F310 in XInput mode, which behaves like an Xbox-style controller and uses familiar A/B/X/Y button names.